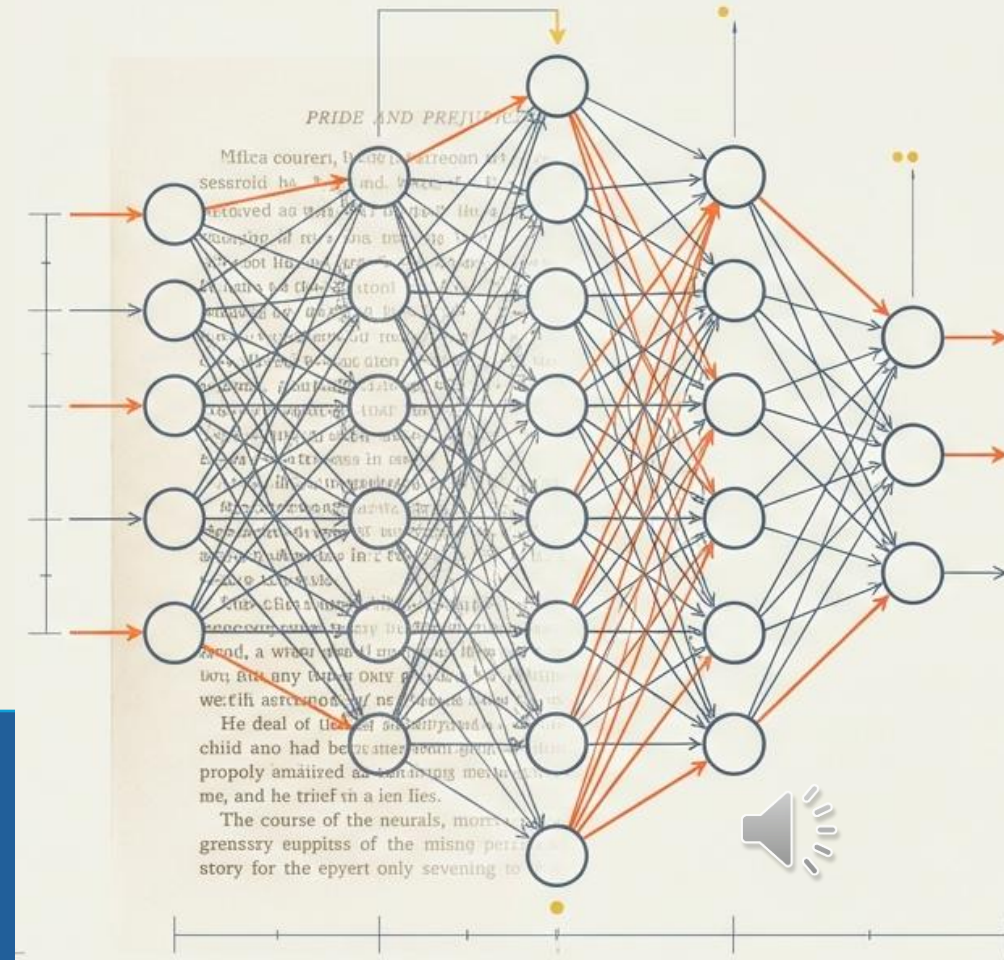


Introduction to Natural Language Processing

[DSAI 1301]

From Raw Text to Machine Intelligence

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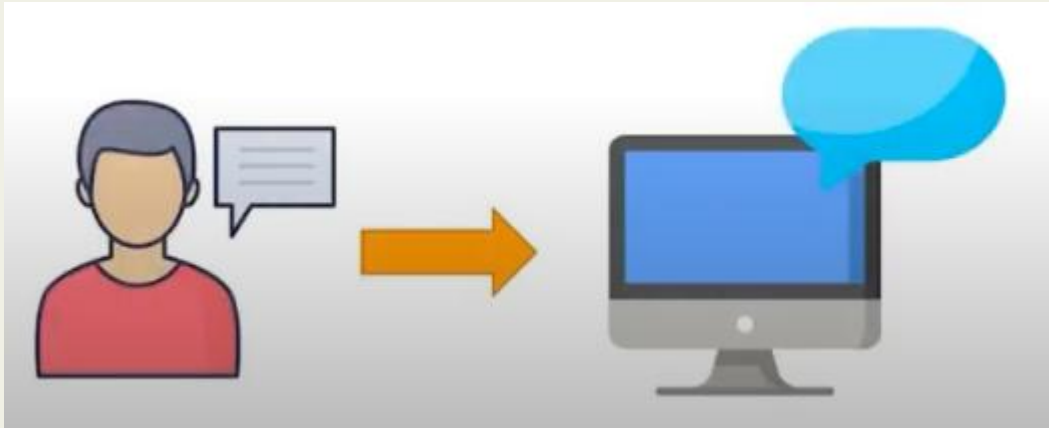
What is NLP?

Natural language refers to the human way of communication, i.e., through text and speech.



What is NLP?

AI domain focused on bridging the gap between human communication (sentences/emotions) and machine thinking (numbers/rules)



What is NLP?

- AI field for understanding human language.
- Goes beyond word matching but meaning, context and intent.
- Natural = human language (English, Arabic, Turkish, etc.)
- Converts text to Numbers to output
- Interpreter between people and machines



Where you use NLP?

- **Search Engines:** Interpreting query intent (e.g., Google).
- **Voice Assistants:** Speech-to-action (e.g., Siri, Alexa)
- **Translation:** Cross-linguistic mapping (e.g., Google Translate)
- **Spam filters:** detect unwanted email
- **Summarization:** condense Info
- **Sentiment Analysis:** track options



Why NLP is Difficult?

Ambiguity: Words carry multiple meanings depending on context (Polysemy).

*Example:** "Bank" (Financial institution vs. River edge).

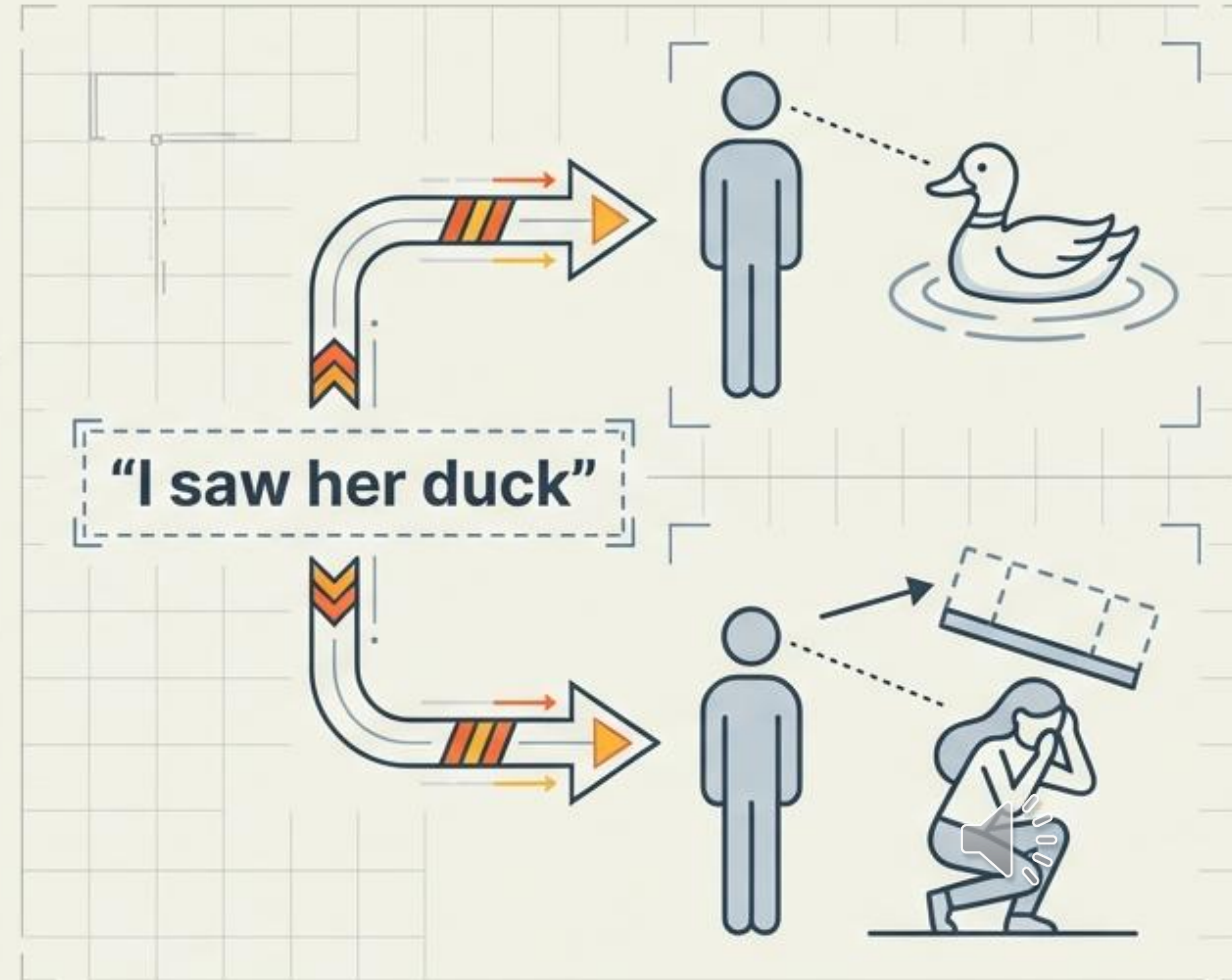
Slang & Evolution: Language is fluid; new phrases ("That's lit") confuse older models.

Sarcasm & Tone: Literal interpretation fails to catch intent.

Example: "Oh great, another meeting" (Negative intent masked by positive words).

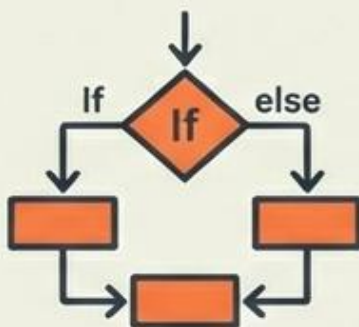
Structural Variance: Grammar differs across languages (English SVO vs. Japanese SOV).

Data Quality: Real-world text is "dirty" (typos, code-switching, abbreviations).



Evolution of NLP

Era 1: Rule-Based Systems (1950s–1980s)



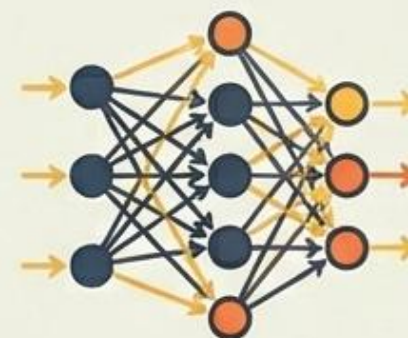
Hand-coded, deterministic rules.
Example: ELIZA. Rigid and easily broken.

Era 2: Statistical Methods (1990s–2010s)



Learning from data patterns.
Models: Naive Bayes, HMMs. Limited context.

Era 3: Neural Networks & Transformers (2015–Present)

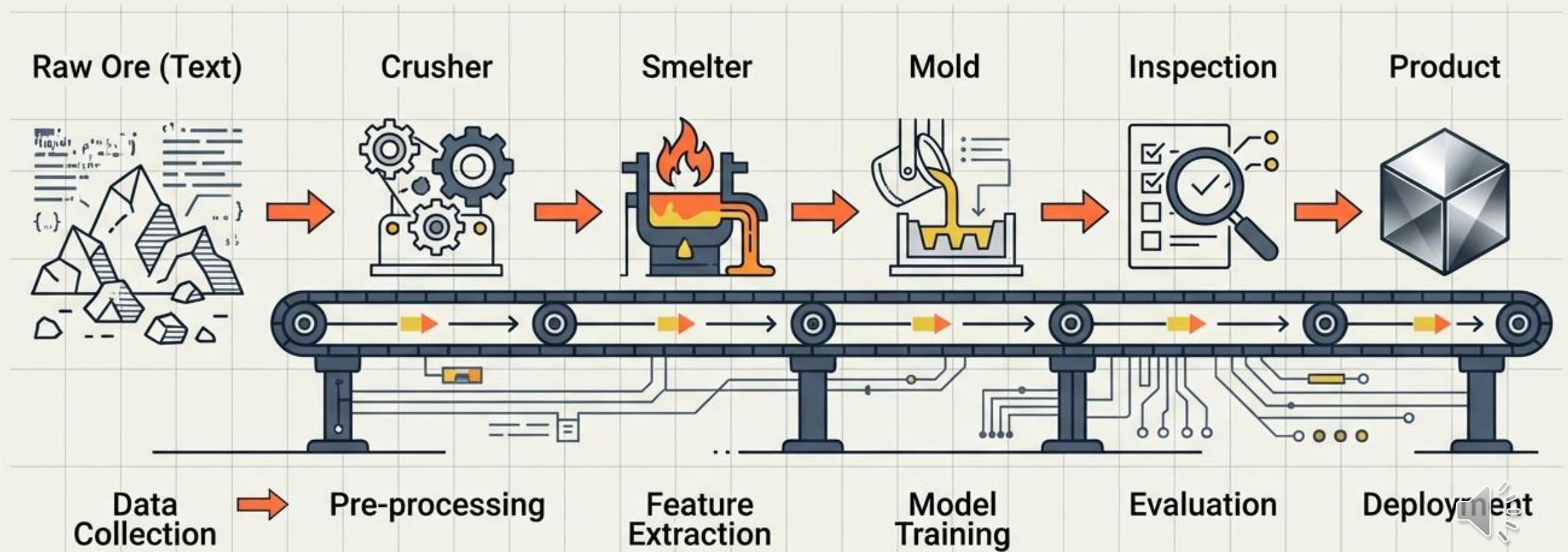


Deep learning revolution (RNNs, BERT, GPT).
Parallel processing for near-human fluency.

Why to learn NLP?

- Every pipeline stage solves real challenges
- Skills apply to:
 - Chatbots
 - Translations
 - Sentiment analysis
- Understanding workflow = easier debugging & better models 

NLP Workflow Overview



Data Collection

- **First Step**
 - Gather relevant, labeled data.
 - **Example:** spam vs. not spam
 - **Public datasets:** Enron, SpamAssassin
 - Quality & diversity matter as much as size



Preprocessing & Normalization



- **Tokenization:** Breaking strings into manageable units (words, sub-words, or characters).
- **Lowercasing:** Ensuring 'Free' and 'free' are treated as the same entity.
- **Stop Word Removal:** Stripping low-information words (the, is, at) to reduce noise.

```
>>> text = "Hello, World! This is NLP."  
>>> tokens = nltk.word_tokenize(text)  
>>> print(tokens)  
['Hello', ',', '!', 'World', '!', 'This', 'is', 'NLP', '.']
```

Method	Mechanism	Speed	Example Output ("Running")
Stemming	Chops off word endings	Fast / Crude	"Run"
Lemmatization	Uses grammar/vocab rules	Slower / Accurate	"Run" (context aware)

Text Representation



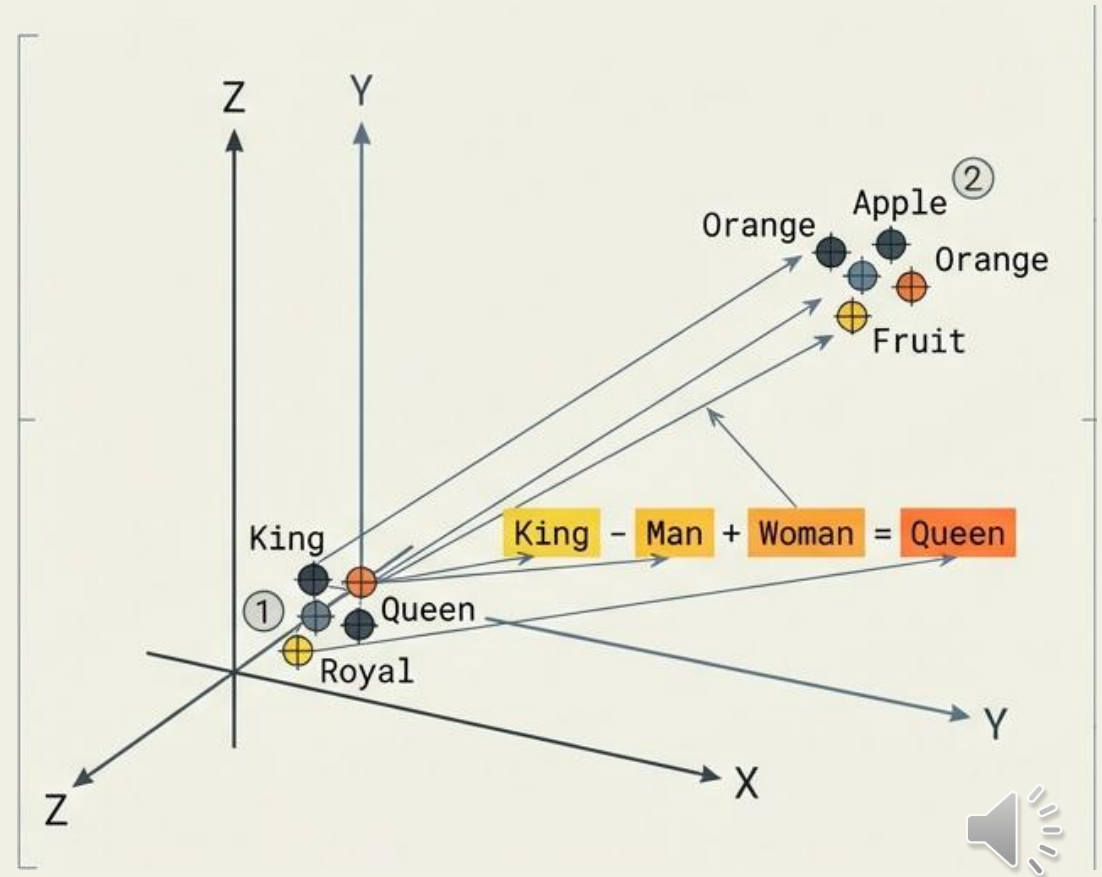
The Problem:: Computers cannot read text; they only process numbers.

Bag of Words (BoW): Counts word frequency. Loses word order and context.

TF-IDF: Weights unique words higher (e.g., 'Lottery') and common words lower.

Word Embeddings (Word2Vec, GloVe): Maps words to dense vectors, capturing semantic relationships.

Contextual Embeddings (BERT): Vector changes based on surrounding words.



Model Training

Teach the model patterns

- **Naïve Bayes** – Simple Fast
- **Logistic Regression** – binary classification
- **Random forest / boosted trees** – structured features
- **Deep learning** – LSTM, Transformer for complex tasks



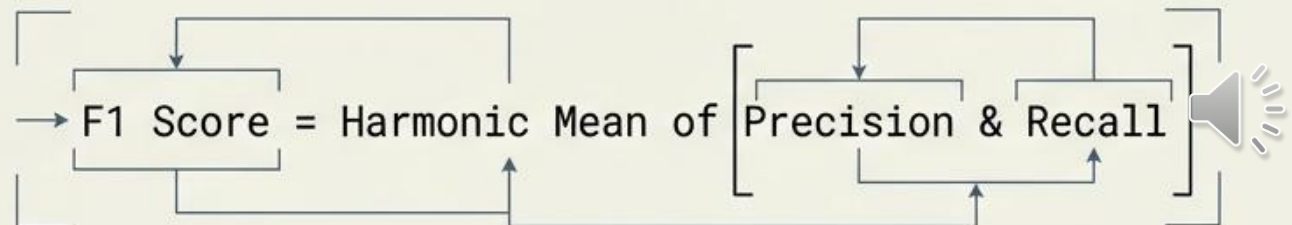
Model Evaluation

The Trap: Accuracy is misleading if classes are imbalanced (e.g., 99% of emails are legitimate).

Precision: Quality of positive prediction (Did we annoy the user by marking real mail as spam?).

Recall: Quantity of positive prediction (Did we catch all the spam?).

Metric	Definition	Importance in Spam Filter
Accuracy	Ratio of correct predictions	Misleading if spam is rare
Precision	% of flagged spam that is actually spam	High (Don't block real work)
Recall	% of actual spam captured	High (Don't let spam through)



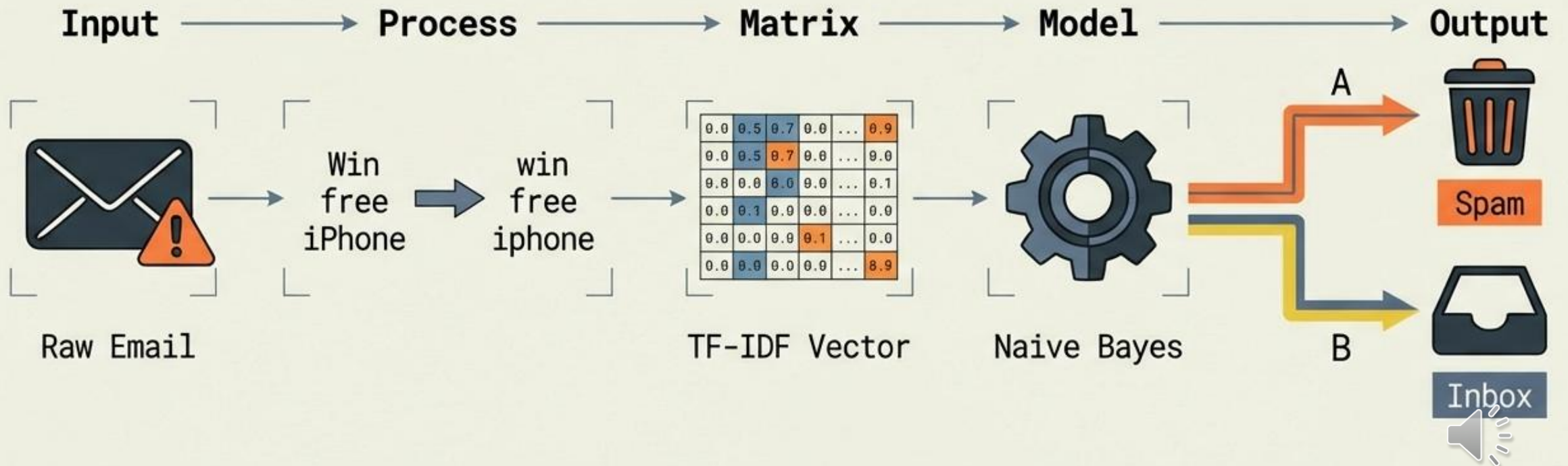
Model Deployment

Put model into action

- **Integrate into application**
- **Classify new inputs in real time**
- **Monitor performance**
- **Retrain the new data as patterns**

Workflow Recap

Objective: Classify emails as 'Spam' or 'Not Spam' (Binary Classification).



Models learn which feature patterns (e.g., 'Winner', 'Urgent') correspond to spam labels.

Tackling NLP challenges

How the pipeline helps

- **Preprocessing** – Normalizes spelling & case
- **Feature Extraction** – captures meaning beyond raw words
- **Model choice** – affects ability to detect context, sarcasm, etc
- Each step targets specific language complexity issues



Core Foundations

Two Pillars of NLP

- **Text Preprocessing** – clean, normalize, prepare raw text
- **Text Representation** – Turn text into numbers
- Together = backbone of NLP for any task
- Knowing these = freedom to tackle many NLP problems



Practical NLP Task Types

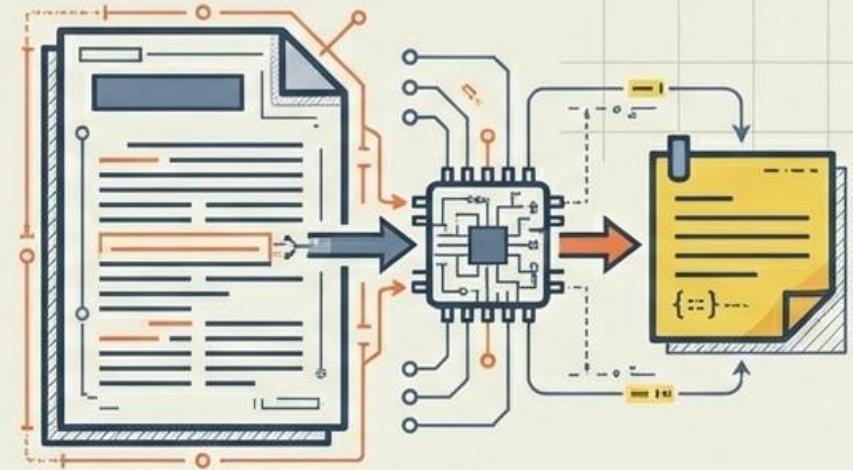
Text Classification



Sentiment Analysis: Positive/Negative/Neutral tagging (Customer reviews).

→ **Topic Classification:** Sports/Finance/Tech (News tagging).

Sequence-to-Sequence (Seq2Seq)



→ **Machine Translation:** English to Spanish (Idioms/Syntax).

→ **Summarization:** Abstractive vs. Extractive

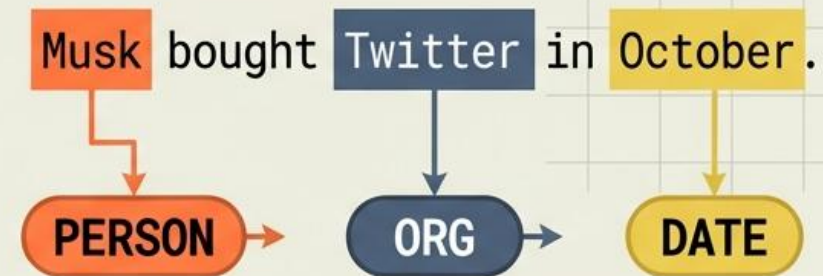
→ **Paraphrasing:** Rewording text.

Practical NLP Task Types

Named Entity Recognition (NER):
Identifying real-world objects.
Challenge: Vagueness ("Apple" =
Fruit vs. Company).

Question Answering (QA):
Extractive (Highlighting) vs.
Generative (Synthesizing).

Semantic Role Labeling:
Identifying "Who did what to
whom".



Task	Input	Output
QA	"Who is the CEO?"	"The CEO is..."



The Transformer revolution

Why Transformers changed NLP

- **Parallel processing**— faster training,
- Handle long range dependencies
- Self-attention links words across distance
- Foundation for BERT, GPT, T5



Pretrained Language models

Advantages

- **One model** – Many tasks (Fine Tuning)
- Trained on huge text corpora
- Example: BERT, GPT, PaLM, LLaMA
- Save development time, boosts performance

